

LABSHEET - 4**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visualization style
sns.set_theme(style="whitegrid")
```

```
# 1. CREATE / LOAD DATASET
# =====

print("=" * 60)
print("EXPERIMENT NO. 4")
print("ADVANCED DATA VISUALIZATION")
print("=" * 60)

# Sample Superstore dataset
data = {
    "Region": [
        "East", "West", "Central", "South", "East",
        "West", "Central", "South", "East", "West",
        "Central", "South", "East", "West", "Central"
    ],
    "Category": [
        "Technology", "Furniture", "Office Supplies", "Technology",
        "Furniture", "Office Supplies", "Technology", "Furniture",
        "Office Supplies", "Technology", "Furniture", "Office Supplies",
        "Technology", "Furniture", "Office Supplies"
    ],
    "Sales": [
        2500, 1800, 1200, 3200, 2100,
        1500, 2800, 1700, 1100, 3500,
        2200, 1300, 2900, 1900, 1400
    ],
    "Profit": [
        500, 300, 180, 700, 350,
        250, 600, 280, 160, 800,
        400, 200, 650, 320, 220
    ],
    "Quantity": [
        10, 7, 5, 12, 8,
        6, 11, 7, 4, 14,
        9, 5, 10, 8, 6
    ],
    "Customer_Age": [
        25, 32, 28, 45, 36,
        29, 41, 38, 24, 50,
        34, 27, 43, 31, 26
    ]
}

df = pd.DataFrame(data)

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Information:")
print(df.info())

print("\nBasic Statistics:")
print(df.describe())
```

```
print(df.describe())
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
EXPERIMENT NO. 4
ADVANCED DATA VISUALIZATION
=====
```

First 5 rows:

	Region	Category	Sales	Profit	Quantity	Customer_Age
0	East	Technology	2500	500	10	25
1	West	Furniture	1800	300	7	32
2	Central	Office Supplies	1200	180	5	28
3	South	Technology	3200	700	12	45
4	East	Furniture	2100	350	8	36

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 15 entries, 0 to 14
```

```
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	Region	15 non-null	object
1	Category	15 non-null	object
2	Sales	15 non-null	int64
3	Profit	15 non-null	int64
4	Quantity	15 non-null	int64
5	Customer_Age	15 non-null	int64

```
dtypes: int64(4), object(2)
```

```
memory usage: 852.0+ bytes
```

```
None
```

Basic Statistics:

	Sales	Profit	Quantity	Customer_Age
count	15.000000	15.000000	15.000000	15.000000
mean	2073.333333	394.000000	8.133333	33.933333
std	758.256145	206.356142	2.850230	7.995237
min	1100.000000	160.000000	4.000000	24.000000
25%	1450.000000	235.000000	6.000000	27.500000
50%	1900.000000	320.000000	8.000000	32.000000
75%	2650.000000	550.000000	10.000000	39.500000
max	3500.000000	800.000000	14.000000	50.000000

```
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```

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```

```
# 2. BAR CHART
```

```
# Compare sales across regions
```

```
# =====
```

```
print("\n" + "=" * 60)
```

```
print("2. BAR CHART - SALES BY REGION")
```

```
print("=" * 60)
```

```
region_sales = df.groupby("Region")["Sales"].sum()
```

```
print("\nSales by Region:")
```

```
print(region_sales)
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(
    region_sales.index,
    region_sales.values
)
```

```
plt.title("Total Sales by Region")
```

```
plt.xlabel("Region")
```

```
plt.ylabel("Total Sales")
```

```
plt.grid(axis="y", alpha=0.3)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
print("Himani Chaudhary")
```

```
print("CU240250967")
```

2. BAR CHART - SALES BY REGION

Sales by Region:

Region

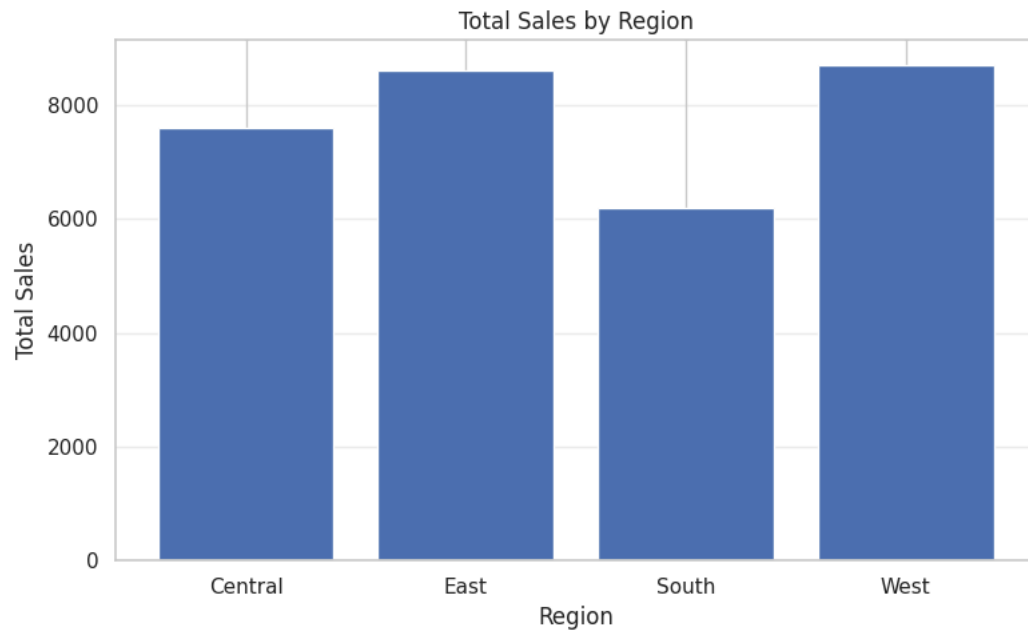
Central 7600

East 8600

South 6200

West 8700

Name: Sales, dtype: int64



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```
# 3. LINE CHART
# Analyze sales trend
# =====

print("\n" + "=" * 60)
print("3. LINE CHART - SALES TREND")
print("=" * 60)

df["Month"] = range(1, len(df) + 1)

plt.figure(figsize=(9, 5))

plt.plot(
    df["Month"],
    df["Sales"],
    marker="o",
    linewidth=2,
    label="Sales"
)

plt.title("Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.legend()

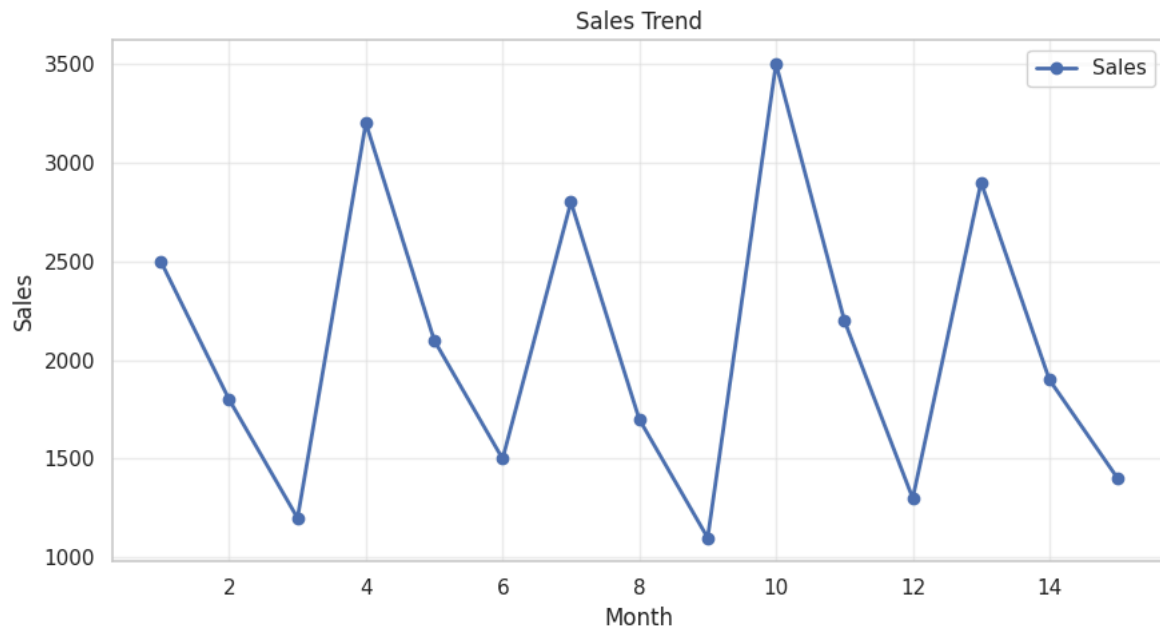
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```

=====

3. LINE CHART - SALES TREND

=====



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```
# 4. HISTOGRAM
# Distribution of sales
# =====

print("\n" + "=" * 60)
print("4. HISTOGRAM - SALES DISTRIBUTION")
print("=" * 60)

plt.figure(figsize=(8, 5))

plt.hist(
    df["Sales"],
    bins=6,
    edgecolor="black"
)

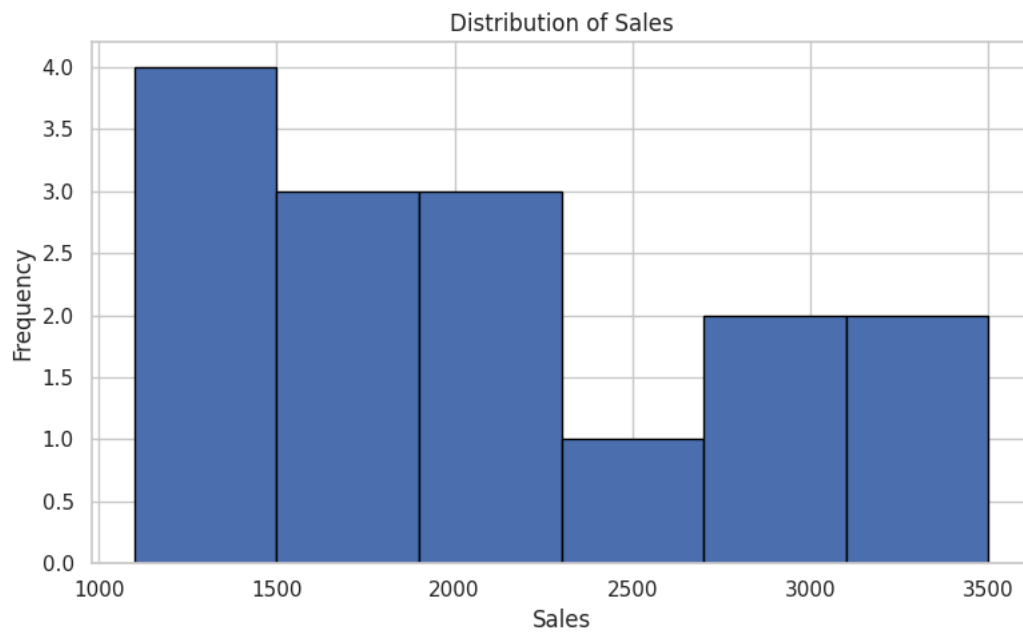
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```

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4. HISTOGRAM - SALES DISTRIBUTION

=====



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```
# 5. BOX PLOT
# Identify outliers and distribution
# =====

print("\n" + "=" * 60)
print("5. BOX PLOT - SALES BY REGION")
print("=" * 60)

plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="Region",
    y="Sales"
)

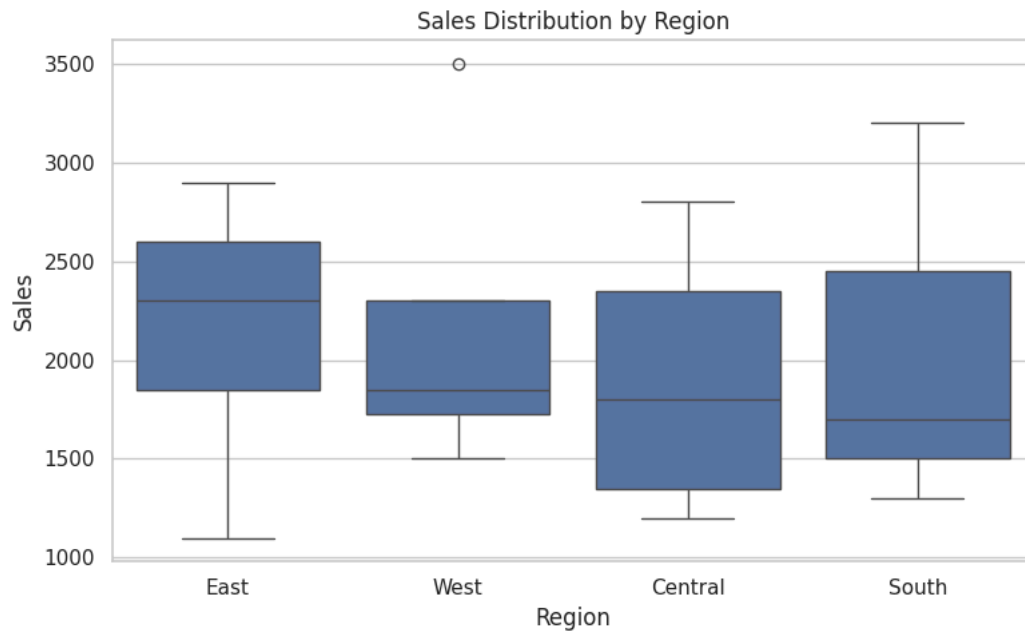
plt.title("Sales Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```

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5. BOX PLOT - SALES BY REGION

=====



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```
# 6. SCATTER PLOT
# Relationship between Sales and Profit
# =====

print("\n" + "=" * 60)
print("6. SCATTER PLOT - SALES VS PROFIT")
print("=" * 60)

plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    s=100
)

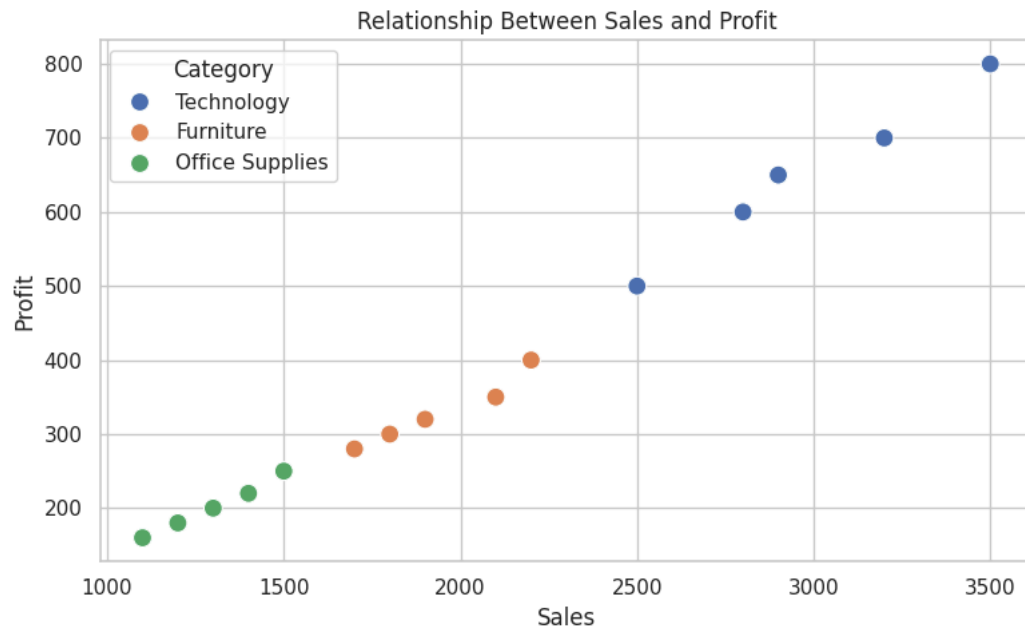
plt.title("Relationship Between Sales and Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.legend(title="Category")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```

=====

6. SCATTER PLOT - SALES VS PROFIT

=====



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```
# 7. CORRELATION HEATMAP
# =====

print("\n" + "=" * 60)
print("7. CORRELATION HEATMAP")
print("=" * 60)

numeric_data = df[
    ["Sales", "Profit", "Quantity", "Customer_Age"]
]

correlation = numeric_data.corr()

print("\nCorrelation Matrix:")
print(correlation)

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

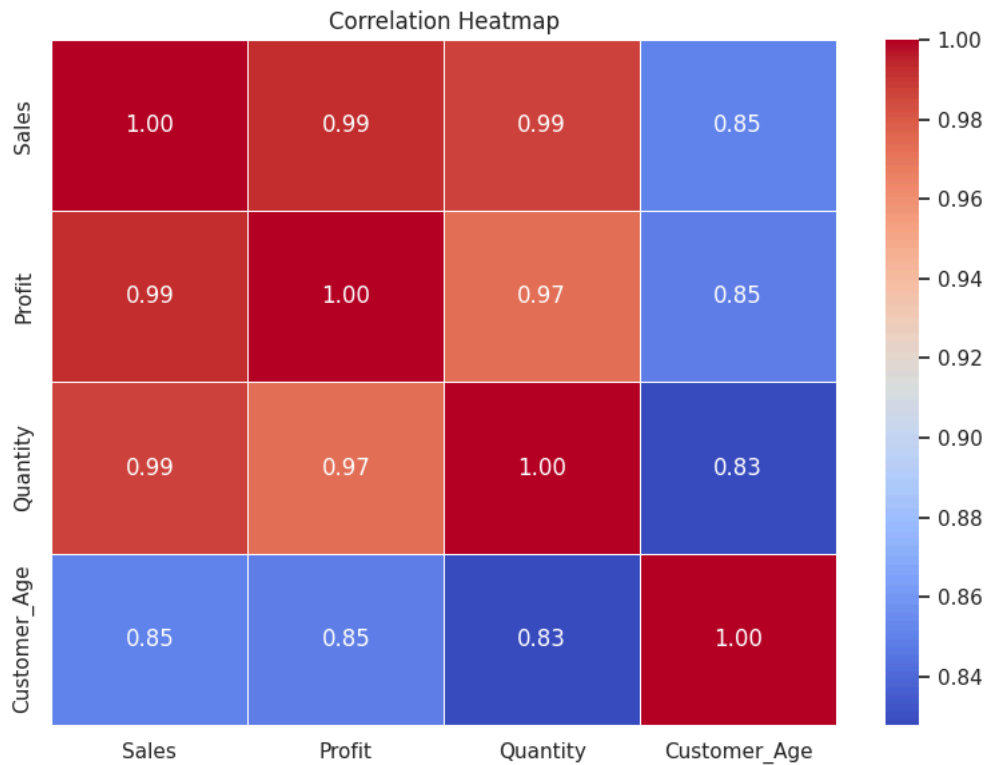
plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

7. CORRELATION HEATMAP

Correlation Matrix:

	Sales	Profit	Quantity	Customer_Age
Sales	1.000000	0.993156	0.986663	0.850357
Profit	0.993156	1.000000	0.973006	0.849160
Quantity	0.986663	0.973006	1.000000	0.827912
Customer_Age	0.850357	0.849160	0.827912	1.000000



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8. SALES BY CATEGORY

```
# =====

print("\n" + "=" * 60)
print("8. SALES BY CATEGORY")
print("=" * 60)

category_sales = df.groupby("Category")["Sales"].sum()

print("\nSales by Category:")
print(category_sales)

plt.figure(figsize=(8, 5))

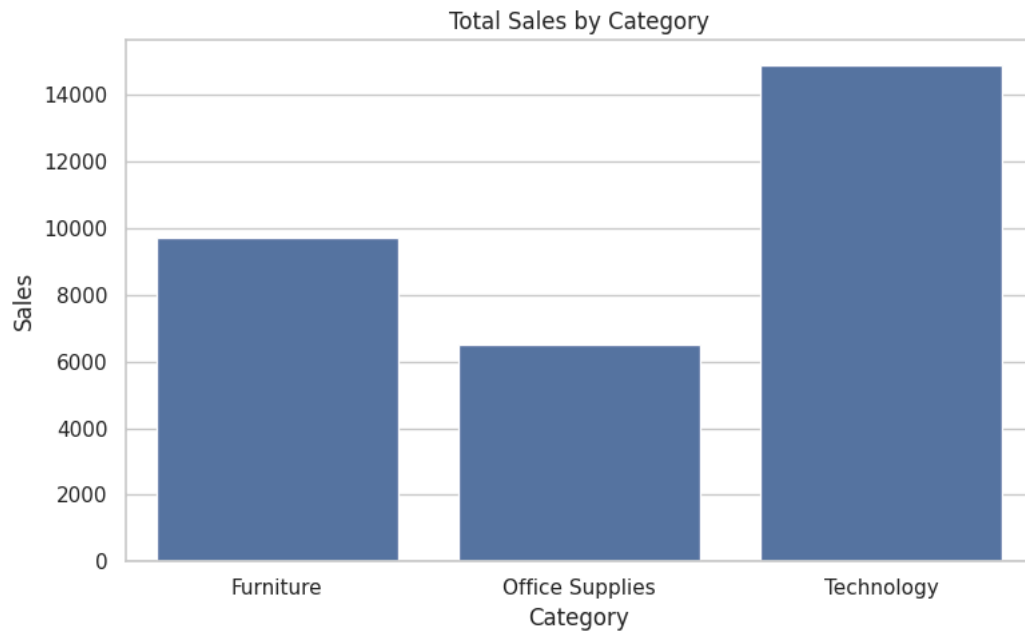
sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```


8. SALES BY CATEGORY

Sales by Category:
 Category
 Furniture 9700
 Office Supplies 6500
 Technology 14900
 Name: Sales, dtype: int64



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```
# 9. PROFIT BY REGION
# =====

print("\n" + "=" * 60)
print("9. PROFIT BY REGION")
print("=" * 60)

region_profit = df.groupby("Region")["Profit"].sum()

print("\nProfit by Region:")
print(region_profit)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=region_profit.index,
    y=region_profit.values
)

plt.title("Total Profit by Region")
plt.xlabel("Region")
plt.ylabel("Profit")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

9. PROFIT BY REGION

Profit by Region:

Region

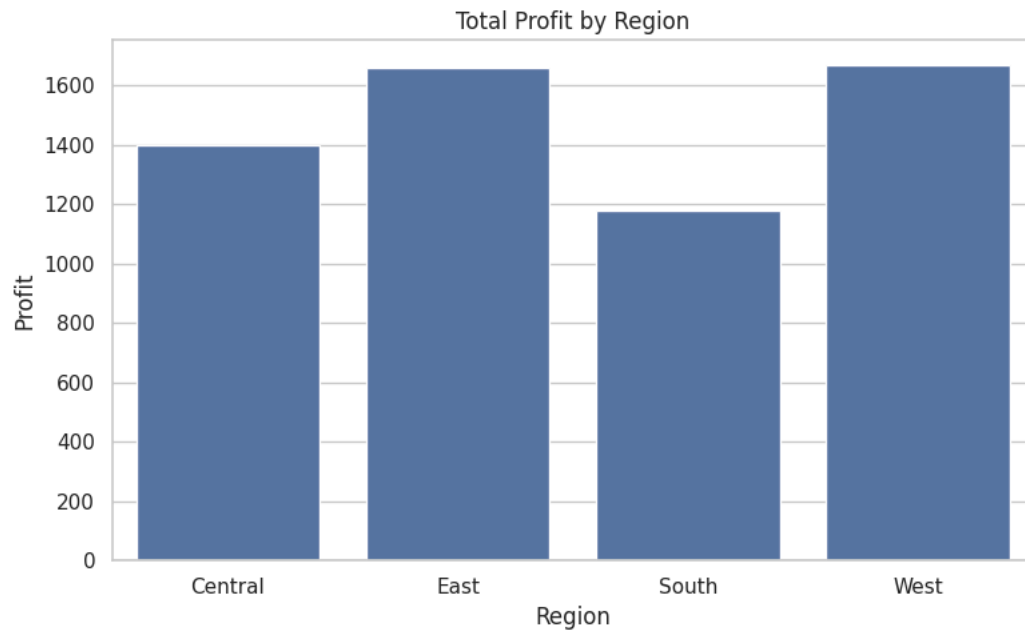
Central 1400

East 1660

South 1180

West 1670

Name: Profit, dtype: int64



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10. MULTIPLE VISUALIZATIONS

```
# =====

print("\n" + "=" * 60)
print("10. ADDITIONAL VISUALIZATION")
print("=" * 60)

plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Customer_Age",
    y="Sales",
    hue="Region",
    s=100
)

plt.title("Customer Age vs Sales")
plt.xlabel("Customer Age")
plt.ylabel("Sales")
plt.legend(title="Region")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

10. ADDITIONAL VISUALIZATION

